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DATASETS, BENCHMARKS, AND PROTOCOLS

Verifying Facts in Patient Care Documents Generated by Large Language Models Using Electronic Health Records

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Abstract

BACKGROUND Large language models (LLMs) can generate clinical documents, but inaccuracies due to hallucinations create a barrier for use in medical practice. Reliance on clinician vigilance to remedy errors is infeasible, since mistakes may be subtle and interleaved among accurate statements within the document. An increasing trend in artificial intelligence (AI) has been to deploy LLMs as evaluators of LLM output (LLM-as-a-Judge), which has been shown to have high concordance with human judgment.

METHODS We hypothesize that an LLM-as-a-Judge with relevant facts from a patient's electronic health record (EHR) can be used to automatically verify facts in any clinical document written about a patient and provide an overall estimate of the document's accuracy. To evaluate this hypothesis, we created a benchmark dataset called VeriFact — Brief Hospital Course (VeriFact-BHC), consisting of statements derived from the brief hospital course sections of discharge summaries, and established a ground truth from clinician chart review. We then devised an AI system, VeriFact, that combines retrieval-augmented generation of facts from a patient's EHR along with an LLM-as-a-Judge to perform automated evaluation of all statements in the dataset.

RESULTS VeriFact-BHC contains 100 patients with 13,070 statements derived from brief hospital courses, each annotated by three or more clinicians. VeriFact achieved 93.2% agreement with clinicians. The highest interrater agreement among clinicians was 88.5%, indicating that VeriFact can produce more consistent fact verification than humans. Scaling the size of the LLM-as-a-Judge model and adding reasoning capability improve different facets of AI judgment and are synergistic when combined.

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CONCLUSIONS VeriFact can help clinicians verify facts in documents drafted by LLMs prior to committing them to the patient's EHR and can automate tasks requiring chart review. VeriFact-BHC can be used to develop and benchmark new methodologies for verifying facts in patient care documents. (Funded by Stanford University and others.)

Introduction

Large language models (LLMs) are effective tools for clinical document generation[1] as they can summarize electronic health records (EHRs),[2,3] perform risk prediction[4] and diagnostic reasoning,[5-7] and translate patient instructions or discharge summaries into patient-friendly language.[8,9] However, LLM-generated documents may contain subtle hallucinations or errors that are difficult to detect, limiting their use.[10-12] Relying on clinician vigilance to detect such errors is unrealistic — akin to asking them to find needles in a haystack.[13]

Similar to how a spellchecker assists a writer with spelling and grammar, VeriFact is an artificial intelligence (AI) system that checks the veracity of statements within an LLM-generated document.[14] New clinical documents should be internally consistent with information already known about a patient, such as historical notes in the patient's EHR. VeriFact performs patient-specific fact verification by comparing statements in an LLM-generated document against a patient's EHR facts, localizes errors, and describes their underlying causes (Fig. 1).

Existing approaches for reference-based fact-checking use Google search results[15] or Wikipedia[16,17] as a reference, but cannot assess statements relative to a patient's unique history. VeriFact is also distinct from reference-free fact-checking approaches such as GPTScore[18] and G-Eval[19] which use a LLM as a text evaluator (LLM-as-a-Judge) and relies on the internal knowledge of the LLM to perform evaluation on arbitrary dimensions such as fluency or coherence. In contrast, VeriFact grounds its evaluation in each patient's EHR, enabling individualized, reference-based fact verification in medicine.

VeriFact splits long-form text into statements that are individually verified.[15,16,20-25] These statements are propositions[26,27] or hypotheses that need to be factually verified. This contrasts with facts, which are historical statements from the patient's EHR that are assumed to be true. For each proposition, facts are retrieved from the patient's EHR to determine whether the proposition is "supported," "not supported," or "not addressed" (Fig. 2). To benchmark VeriFact against human clinicians, this study introduces a clinician-annotated dataset called VeriFact — Brief Hospital Course (VeriFact-BHC), which comprises brief hospital course (BHC) sections from discharge summaries. Each BHC — a concise summary of the patient's hospitalization, including major events, treatments, and outcomes — is transformed into propositions and independently verified by clinicians through chart review.

This study evaluates VeriFact's accuracy by measuring human–AI agreement across various proposition and fact representations, the amount of information retrieved from the EHR, and the effect of increasing the model size and reasoning capability[28,29] of the LLM-as-a-Judge in VeriFact.

Methods

This study adheres to the TRIPOD+AI[30] and TRIPOD-LLM[31] statements of the TRIPOD (Transparent Reporting of a multivariable prediction model for Individual Prognosis or Diagnosis) reporting guidelines. All data were deidentified and the Stanford University Institutional Review Board determined that the study did not constitute human subjects research.

DATASET CREATION

The VeriFact-BHC dataset[32] used in this study was derived from the Medical Information Mart for Intensive Care (MIMIC)-III Clinical Database v1.4, which contains clinical notes from patients who received care at the tertiary/quaternary care hospital Beth Israel Deaconess Medical Center in Boston, Massachusetts, USA.[33] MIMIC-III was chosen for its longitudinal notes, which resemble a production EHR, instead of MIMIC-IV, which includes only radiology and discharge notes. A total of 100 patients was sampled based on the criteria outlined in Figure S1 in the Supplementary Appendix. The original human-written BHC was extracted from the discharge summary, and the remaining notes collectively formed the patient's reference EHR. Llama-3.1-70B-Instruct[34] was used to summarize all EHR notes from the most recent hospital admission to create an LLM-generated BHC using prompts detailed in Figure S2. To promote diversity in writing style, the LLM temperature was set to 0.5.

Each human- and LLM-written BHC was decomposed into propositions using two methods. Sentence propositions were obtained by splitting each BHC using the Natural Language Toolkit[35] sentence parser. Atomic claim propositions[16,24,36] were produced by prompting

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A LLM written "Brief Hospital Course"

The patient, a 77-year-old male, was admitted with a diagnosis of acute cholangitis, presenting with nausea, vomiting, and abdominal pain, and a history of coronary artery disease, atrial fibrillation, cardiomyopathy, and chronic obstructive pulmonary disease. Initial imaging revealed gallstones, and the patient underwent a series of interventions, including fluid boluses, IV antibiotics, and a central line placement. Due to worsening septic shock, the patient required vasopressors and was intubated. An ERCP procedure was performed, which revealed pus, sludge, and stones in the common bile duct, and a stent was placed for drainage. Post-procedure, the patient's condition improved, with stabilization of blood pressure, improvement in abdominal pain, and eventual weaning off vasopressors. The patient was extubated and transferred to the floor, where he continued to receive treatment for sepsis, acute renal failure, and pain control, with plans for repeat ERCP in 4 weeks.

B VeriFact Evaluation Score Sheet

90% Supported (27 Propositions)
The text was supported by the reference context for various reasons including the performance and findings of an ERCP procedure, the patient's condition and treatment post-procedure, the patient's medical history, and the patient's symptoms and treatment during hospitalization. The reference context included multiple sources such as nursing progress notes, physician attending progress notes, radiology notes, and more, which all supported the information presented in the text.

7% Not Supported (2 Propositions)
The text was not supported by the reference context due to two main reasons: (1) the patient's history of cardiomyopathy was not mentioned in the reference context, despite a history of extensive cardiac issues, and (2) the reason for intubation was incorrectly stated as worsening septic shock, when in fact the patient was intubated for procedures and was later extubated.

3% Not Addressed (1 Proposition)
The reference context does not mention the patient presenting with vomiting.

C Detailed Proposition Level Explanations

Proposition	EHR Facts Reference Context	Verdict and Reason
The patient had a history of chronic obstructive pulmonary disease.	Hospital Admission Start: September 18, 2151 03.04 p.m., Hospital Admission End: September 23, 2151 03.30 p.m., Electronic Health Record Context Ordered by Time (Earliest to Latest): 1. Date: September 18, 2151, Time: 08:25 p.m., Note Category: Physician, Note Description: Physician Resident Admission Note | Text: The patient has a past medical history of coronary artery disease. 2. Date: September 18, 2151, Time: 08:25 p.m., Note Category: Physician, Note Description: Physician Resident Admission Note | Text: **The patient has Chronic Obstructive Pulmonary Disease.** *... <23 more EHR claims> ...*	**Supported** The text states that the patient had a history of chronic obstructive pulmonary disease, which is supported by multiple entries in the reference context, including entries from both nursing and physician notes.
The patient was intubated due to worsening septic shock.	*... <Hospital Admission Info + 8 EHR Claims> ...* 9. Date: September 19, 2151, Time: 09:51 a.m., Note Category: Nursing, Note Description: Nursing Progress Note | Text: **The patient was intubated for the procedure.** 10. Date: September 19, 2151, Time: 09:51 a.m., Note Category: Nursing, Note Description: Nursing Progress Note | Text: **The patient returned to MICU intubated.** 11. Date: September 19, 2151, Time: 09:51 a.m., Note Category: Nursing, Note Description: Nursing Progress Note | Text: **The patient was extubated shortly after return to MICU.** *... <14 more EHR claims> ...*	**Not Supported** The text states that the patient was intubated due to worsening septic shock, but the reference context shows that the patient was intubated for procedures and was extubated shortly after return to MICU.

Figure 1. VeriFact Evaluation of a Large Language Model–Generated Clinical Narrative.

Illustration of the evaluation of a large language model (LLM)–generated clinical narrative by the VeriFact system. Panel A shows an LLM-generated brief hospital course narrative that summarizes the hospital admission, created by applying iterative rolling summarization across all of the patient's electronic health record (EHR) clinical notes from the admission. Panel B shows the VeriFact evaluation score sheet output when verifying the LLM-generated brief hospital course with respect to the patient's EHR clinical notes. The score sheet provides an overview of how much of the LLM-generated brief hospital course text is supported by the patient's EHR and outputs human-readable explanations for the scores. The explanations are generated by summarizing the reasons for all propositions classified as "supported," "not supported," and "not addressed," respectively. Panel C shows an illustration of each proposition and the corresponding reference context of EHR facts being presented to the LLM as an evaluator of LLM output (LLM-as-a-Judge), along with the verdict and explanation assigned by the LLM-as-Judge evaluator. EHR denotes electronic health record; ERCP, endoscopic retrograde cholangiopancreatography; IV, intravenous; LLM, large language model; and MICU, medical intensive care unit.

Llama-3.1-70B-Instruct (temperature=0.1) to rewrite the document as a set of subject–object–predicate statements (Fig. S3). Although sentences may cover multiple topics, atomic claims isolate each topic as a basic, indivisible statement (Fig. S4). Full details for dataset creation are provided in the Methods section of the Supplementary Appendix.

DATA ANNOTATION

VeriFact-BHC was prospectively annotated to establish a human clinician ground truth for benchmarking VeriFact's performance on fact verification. Each proposition was annotated as supported, not supported, or not addressed. Each clinician annotator received a PDF of the patient's

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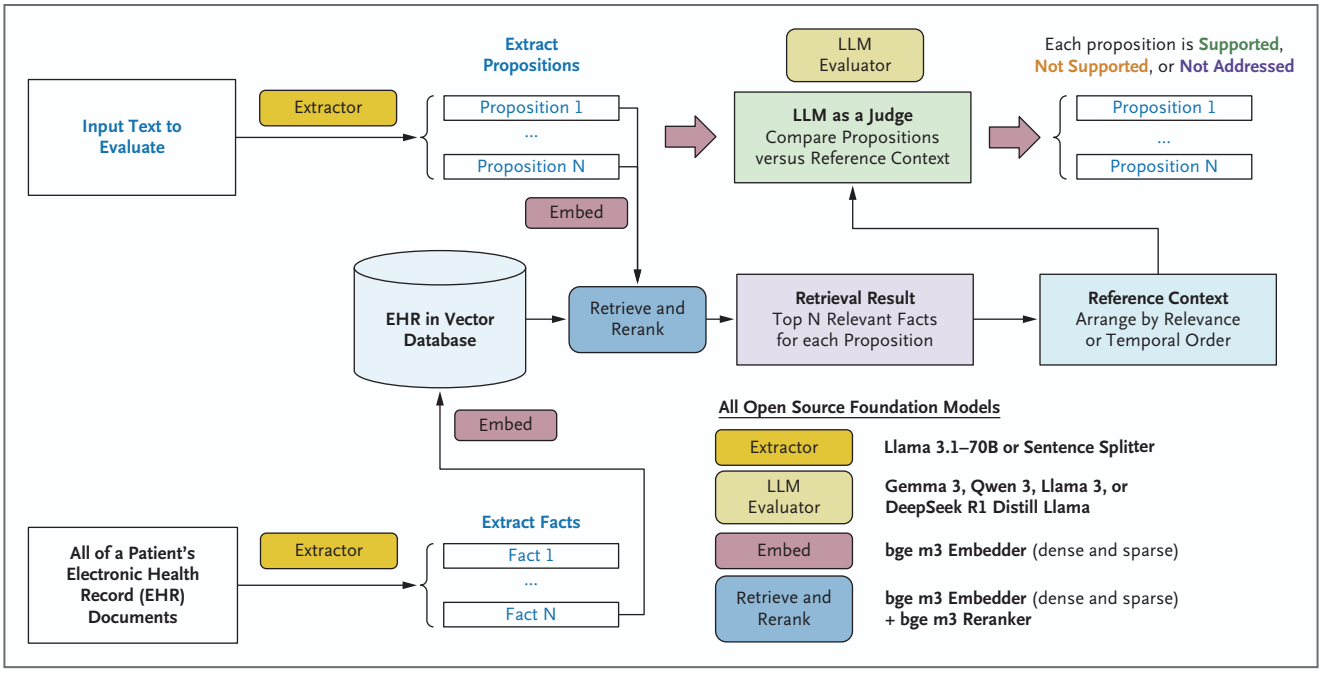


Figure 2. VeriFact System Architecture.

Schematic of the VeriFact system. VeriFact employs a large language model (LLM) to transform long-form input text, such as brief hospital course narratives, into a set of propositions for more detailed evaluation. For each proposition, VeriFact dynamically retrieves only the most relevant facts in the electronic health record (EHR) to form a reference context-specific to that proposition. Each proposition and its corresponding reference context of facts are presented to an LLM as an evaluator of LLM output (LLM-as-a-Judge) tasked with determining whether each proposition is "supported," "not supported," or "not addressed" by the facts in the EHR. Multiple foundation models are used in VeriFact to perform different functions: (1) extractor refers to either a sentence splitter for extracting sentence propositions and facts or a Llama-3.1-70B model for extracting atomic claim propositions and facts; (2) embed refers to the BAAI/bge-m3 biencoder language model that produces vector representations of propositions and facts for vector search, in which dense vector search utilizes semantic similarity, whereas sparse vectors search utilizes token and lexical similarity to perform information retrieval; (3) rerank refers to the BAAI/bge-reranker-v2-m3 cross-encoder language model that can jointly attend to both proposition and fact and assign a more nuanced ranking score than biencoder language models; and (4) LLM evaluator refers to an LLM prompted with the evaluation task of judging whether a proposition is supported by the reference context of EHR facts. Models studied for LLM-as-a-Judge include models from the Gemma-3, Qwen-3, Llama-3, and DeepSeek-R1-Distill-Llama families. Experiments are conducted using only open-source foundation models to maximize transparency and reproducibility, but similar private or close-sourced foundation models can be substituted. bge-m3 denotes BAAI/bge-m3; EHR, electronic health record; LLM, large language model; and LLM-as-a-Judge, LLM as an evaluator of LLM output.

predischarge EHR and could search the text for information (Fig. S5). Annotators were informed that each proposition was derived from a BHC discharge summary and should be interpreted as reflecting the clinical state at the time the discharge note was written. In the first round of annotation, each proposition was assigned a label by 3 of 25 clinicians, and a majority vote was taken to mitigate individual annotator biases. Propositions with three-way disagreements were independently annotated by two new physicians. If both agreed, their labels were accepted; otherwise, disagreements were adjudicated via group discussion with a third physician to break ties until a consensus ground truth label was established. Clinician annotators were blinded to whether the proposition originated from a human-written or an LLM-generated BHC. Annotation was conducted using a custom annotation interface in Label Studio (Fig. S5).[37]

THE VERIFACT SYSTEM

VeriFact combines retrieval-augmented generation with LLM-as-a-Judge (Fig. 2). Retrieval-augmented generation provides LLMs with relevant contextual information in the prompt and is used to dynamically approximate the relevant

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portions of the patient's EHR. LLM-as-a-Judge utilizes LLMs as an AI evaluator of LLM output. VeriFact comprises a vector database with three types of foundation models: an LLM, a text embedding model, and a text reranking model. To study the models' generalization capabilities, none of the models was trained on any data or tasks in this study.

All EHR documents were decomposed into atomic claims and sentence propositions and indexed in a vector database with dense and sparse text embeddings produced by BAAI/bge-m3, a multilingual embedding model developed by the Beijing Academy of Artificial Intelligence for high-accuracy text retrieval and semantic similarity tasks. For the evaluation of clinical text such as BHCs, VeriFact transforms the text into a set of propositions, then retrieves the top *N* most relevant EHR facts from the vector database and formats them into a reference context to approximate all relevant knowledge needed to evaluate the proposition.[38] Both proposition and reference context are placed in a prompt for the LLM-as-a-Judge[39,40] to classify the proposition as supported, not supported, or not addressed by the patient's EHR and provide a reason for its classification (Fig. S6). When all propositions are evaluated, the percentage of supported, not supported, and not addressed labels is determined, and all verdict explanations are summarized for each label category (Fig. 1).

The EHR fact retrieval step involves using the BAAI/bge-m3 model to produce semantic (dense)[41] and lexical, word-based (sparse),[42] vector representations of the proposition, which are used to identify relevant facts from the EHR. Hybrid search[43] is employed, which combines dense and sparse search results using distribution-based score fusion,[44] followed by the BAAI/bge-reranker-v2-m3 to ensure that the most relevant facts appear at the top of the search results. Final retrieved facts are ordered chronologically and formatted into a reference context along with the date, time, and note type from which the fact originated.

Multiple LLM judges were studied as evaluators: Gemma-3-12B, Gemma-3-27B, Qwen-3-30B-A3B-Instruct, Qwen-3-30B-A3B-Thinking, Qwen-3-32B, DeepSeek-R1-Distill-Llama-8B, DeepSeek-R1-Distill-Llama-70B, Llama-3.1-8B-Instruct, and Llama-3.1-70B-Instruct.[28,34,45,46] Qwen-3-30B-A3B-Thinking, DeepSeek-R1-Distill-Llama-8B, and DeepSeek-R1-Distill-Llama-70B are reasoning models that produce an intermediate reasoning chain, whereas the other models do not. Full implementation details are provided in the Methods section of the Supplementary Appendix.

STATISTICAL ANALYSIS AND EVALUATION

Inter-rater agreement between clinicians on proposition label assignment on VeriFact-BHC was measured with percent agreement and Gwet's AC1[47] using irrCAC version 0.4.3. Agreement between VeriFact and human clinician ground truth was measured using percentage agreement, Gwet's AC1, and Matthews correlation coefficient (MCC)[48,49] using scikit-learn v1.6.1. Percent agreement averages concordance across all annotation pairs, whereas Gwet's AC1 adjusts for chance agreement while being less sensitive to label imbalance and the kappa paradox.[50] The percent agreement and Gwet's AC1 reflect overall agreement but do not account for per-class performance, whereas a high MCC requires accurate prediction for each and every label class. Confidence intervals (CIs) of 95% were estimated using 1000 bootstrap iterations. Preliminary univariate sensitivity analyses were conducted using Llama-3.1-8B-Instruct, Llama-3.1-70B-Instruct, DeepSeek-R1-Distill-Llama-8B, and DeepSeek-R1-Distill-Llama-70B to identify the optimal retrieval method (details are in the Methods section of the Supplementary Appendix), and these settings were used for the main experiments comparing VeriFact against human ground truth labels.

DATA AND CODE AVAILABILITY

The VeriFact-BHC dataset is publicly available on PhysioNet per the credentialed access policy and data use agreement at https://physionet.org/content/mimic-iii-ext-verifact-bhc/. Code for the VeriFact system, VeriFact-BHC dataset creation, and experimental analysis is publicly available at https://github.com/philipchung/verifact.

Results

DATASET AND ANNOTATION

VeriFact-BHC contains records from 100 patients, each with LLM-generated BHC, human-written BHC, and clinical notes from each patient's EHR (Table S1). The final dataset contains 13,070 propositions (4445 from LLM-generated BHCs and 8625 from human-written BHCs), 9095 atomic claim propositions, and 3975 sentence propositions. Atomic claims enabled finer-grained evaluation of BHC text with an average of 2.2 atomic claims per sentence.

Each proposition was annotated by three or more clinicians across multiple rounds. Of the 478 propositions that had complete disagreement between three clinicians after one round of annotation and underwent additional annotation,

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231 required manual adjudication by three clinicians in a group discussion (Table S4). Annotators collectively logged 1618 hours (67.4 days) of annotation time in Label Studio. Clinicians had 5±9.2 years of practice (mean±standard deviation) and 8±4.5 years of EHR experience and represented anesthesiology, pediatrics, internal medicine, general practice, neurology, emergency medicine, and critical care medical specialties.

CLINICIAN INTER-RATER AGREEMENT

Clinician inter-rater variation reflected differences in chart review practice, judgement thresholds, and interpretations of the patient's EHR data. Clinician annotations for LLM-generated BHCs show that sentence propositions had lower agreement at 84.7% (95% CI, 83.0 to 86.7%) compared with atomic claim propositions at 88.5% (95% CI, 87.6 to 89.4%) (Table 1). This is due to ambiguity arising from forcing a single label assignment to sentence propositions that discussed multiple topics, each of which may or may not be supported by EHR facts. The same pattern is seen in human-written BHCs. Because of this, clinicians have greater consensus for atomic claim propositions, which is desirable when establishing ground truth labels.[51-53] Agreement on the majority supported label is high, but lower when at least one annotator chose not supported or not addressed; this lower agreement persists even after binarizing labels to supported versus not supported or addressed (Table S2), highlighting that disagreement arises primarily in negative cases, which may also pose greater difficulty for automated fact verification.

HUMAN–ARTIFICIAL INTELLIGENCE AGREEMENT

Preliminary sensitivity analysis on several hyperparameters determined that the best VeriFact performance can be achieved using the rerank EHR fact retrieval method, absolute time reference content formatting, and limiting fact retrieval to the current admission; these values were adopted for all subsequent experiments. Full details of the preliminary analyses are in the Results section of the Supplementary Appendix.

Human–AI agreement for VeriFact evaluating LLM-generated BHCs decomposed to atomic claim propositions is shown in Figure 3. The reasoning LLM judge, DeepSeek-R1-Distill-Llama-70B, combined with 100 sentence EHR facts, resulted in the highest agreement with human ground truth at 93.2% (95% CI, 92.3 to 94.0%) while also maintaining the highest per-class performance shown by an MCC of 0.31 (95% CI, 0.24 to 0.35). Top nonreasoning models Llama-3.1-70B, Qwen-3-32B, and Qwen-3-30B-A3B-Instruct all achieve high agreement but have lower MCCs. Qwen-3-30B-A3B-Thinking achieves a higher MCC but with lower

Table 1. Inter-Clinician Agreement on the VeriFact — Brief Hospital Course Dataset.*

	Author Type	LLM		Human	
Proposition Stratification	**Proposition Type**	**Atomic Claim**	**Sentence**	**Atomic Claim**	**Sentence**
All propositions (n=13,070)	Percent agreement (95% CI)	88.5% (87.6 to 89.4%)	84.7% (83.0 to 86.7%)	73.3% (72.3 to 74.2%)	66.9% (65.6 to 68.2%)
	Gwet's AC1 (95% CI)	0.88 (0.87 to 0.89)	0.83 (0.81 to 0.85)	0.64 (0.63 to 0.65)	0.53 (0.51 to 0.55)
Propositions with at least one clinician assigning "supported" (n=10,775)	Percent agreement (95% CI)	88.7% (87.8 to 89.5%)	84.9% (83.1 to 86.7%)	75.6% (74.6 to 76.6%)	66.7% (65.0 to 68.4%)
	Gwet's AC1 (95% CI)	0.88 (0.87 to 0.89)	0.84 (0.81 to 0.86)	0.71 (0.70 to 0.73)	0.59 (0.57 to 0.61)
Propositions with at least one clinician assigning "not supported" (n=3319)	Percent agreement (95% CI)	34.0% (32.1 to 36.1%)	31.4% (29.4 to 33.7%)	33.3% (32.2 to 34.4%)	32.3% (31.0 to 33.6%)
	Gwet's AC1 (95% CI)	0.09 (0.06 to 0.12)	0.06 (0.03 to 0.10)	0.02 (0.00 to 0.03)	0.01 (−0.02 to 0.03)
Propositions with at least one clinician assigning "not addressed" (n=3770)	Percent agreement (95% CI)	31.1% (28.9 to 33.6%)	30.8% (25.6 to 35.9%)	45.8% (44.3 to 47.2%)	49.2% (47.3 to 51.1%)
	Gwet's AC1 (95% CI)	0.03 (−0.01 to 0.07)	0.02 (−0.06 to 0.10)	0.25 (0.23 to 0.28)	0.31 (0.28 to 0.34)

*Clinician interannotator agreement for whether each proposition is "supported," "not supported," or "not addressed" by a patient's electronic health record clinical notes. Agreement is measured with percent agreement and Gwet's AC1 with 95% confidence intervals. Each proposition was labeled by three clinician annotators. In addition to showing agreement across all propositions, the table also shows agreement computed over a subset of propositions where at least one clinician assigned a supported, not supported, or not addressed verdict. These subset analyses reveal that a majority of inter-clinician disagreement occurred for propositions where at least one clinician assigns a not supported or not addressed verdict. CI denotes confidence interval; and LLM, large language model.

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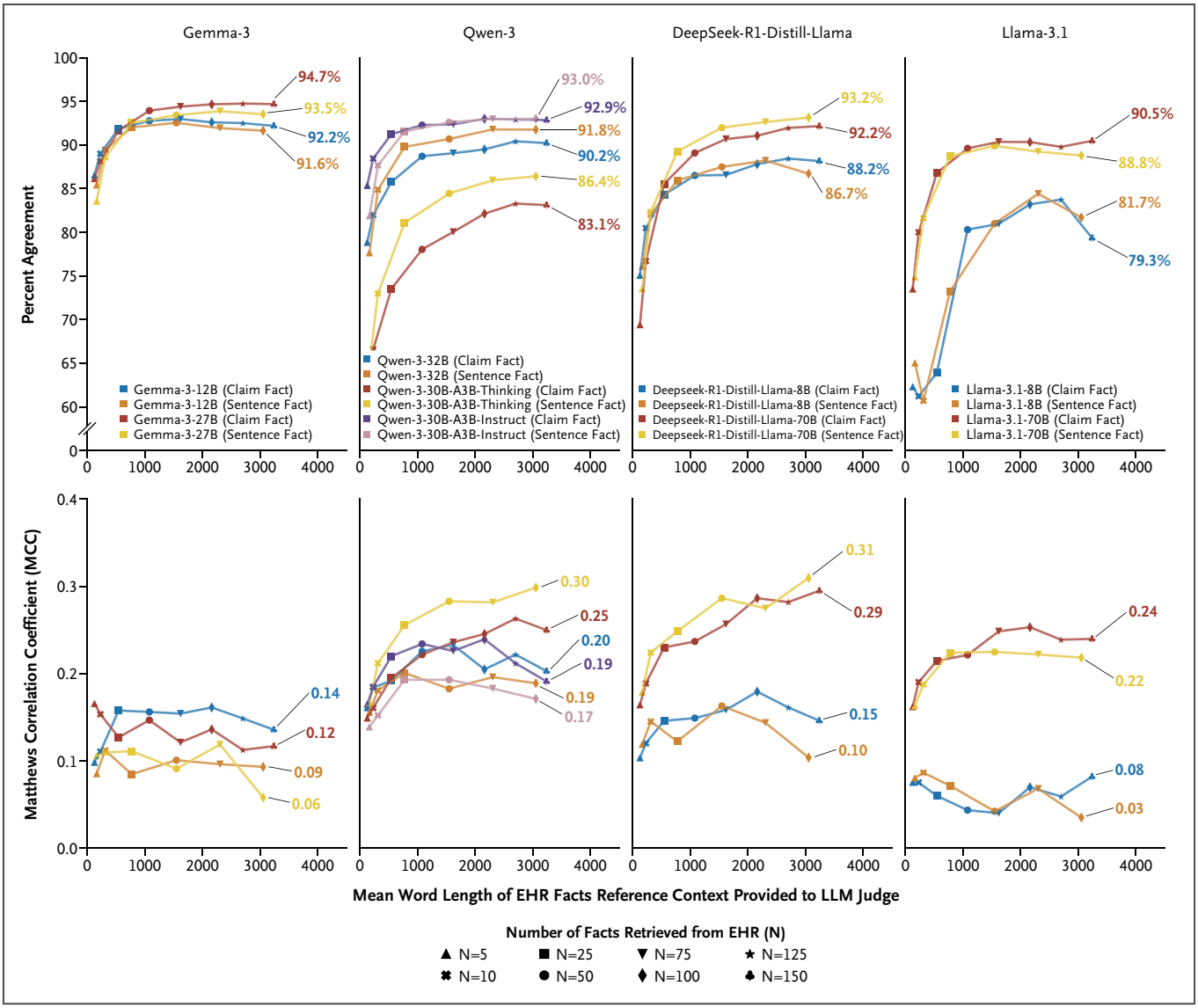


Figure 3. VeriFact Performance for Large Language Model–Generated Brief Hospital Course Narratives Evaluated Using Atomic Claim Propositions.

Percent agreement (top plots) and Matthews correlation coefficient (bottom plots) when comparing VeriFact against human clinician ground truth labels in a three-label classification task ("supported," "not supported," "not addressed"). The evaluation is performed on a large language model–written brief hospital course summaries. The unit of evaluation is atomic claim propositions. The plots show a sensitivity analysis across three aspects of the VeriFact system: (1) increasing the number of retrieved facts from the electronic health record, (2) representation of facts in the electronic health record documents as atomic claim facts versus sentence facts, and (3) using different large language models as evaluators of large language model output. EHR denotes electronic health record; LLM, large language model; and MCC, Matthews correlation coefficient.

agreement because it tends to agree with humans on not supported and not addressed labels at the cost of agreement on supported classifications. Gemma-3 models achieve the highest agreement but have low MCCs and tend to overpredict the majority label supported. These findings are reflected in the confusion matrices depicted in Figure S7.

As a secondary analysis, BHCs were decomposed to sentence propositions rather than atomic claim propositions and then evaluated with VeriFact (Figs. S8 and S9). Many models achieve a percent agreement above 92%, but they all suffer from lower MCCs between 0 and 0.13. Qwen-3-30B-A3B-Thinking, with sentence facts, achieves a higher

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MCC of 0.23 (95% CI, 0.16–0.31) but with lower agreement of 84.9% (95% CI, 82.5–87.2%). These findings show that VeriFact performs worse with sentence propositions due to the ambiguity that can arise when sentences contain multiple facts to evaluate. Experiments were also conducted for human-written BHCs (see the Results in the Supplementary Appendix) with similar findings.

Across all scenarios, reasoning models achieved both high percent agreement and MCCs, whereas nonreasoning models achieved high percent agreement at the cost of lower MCCs. Using atomic claims instead of sentence propositions as a unit of evaluation improves VeriFact agreement, and using sentence EHR facts instead of atomic claim in EHR facts improves VeriFact agreement (Fig. 3, Figs. S8 and S18, and Table S6). This indicates that original source sentences are a better representation of the EHR fact reference context.

The percent agreement, Gwet's AC1, MCC, sensitivity, specificity, positive predictive value, and negative predictive value for each hyperparameter combination tested are reported in Table S3.

IMPACT OF MODEL SIZE AND REASONING BEHAVIOR FOR LARGE LANGUAGE MODELS AS EVALUATORS OF LARGE LANGUAGE MODEL OUTPUT

As the DeepSeek-R1-Distill-Llama family of models are the Llama-3 models with additional reasoning training, a direct comparison can be made to examine the effect of model size and reasoning (Fig. 4). Both increases in

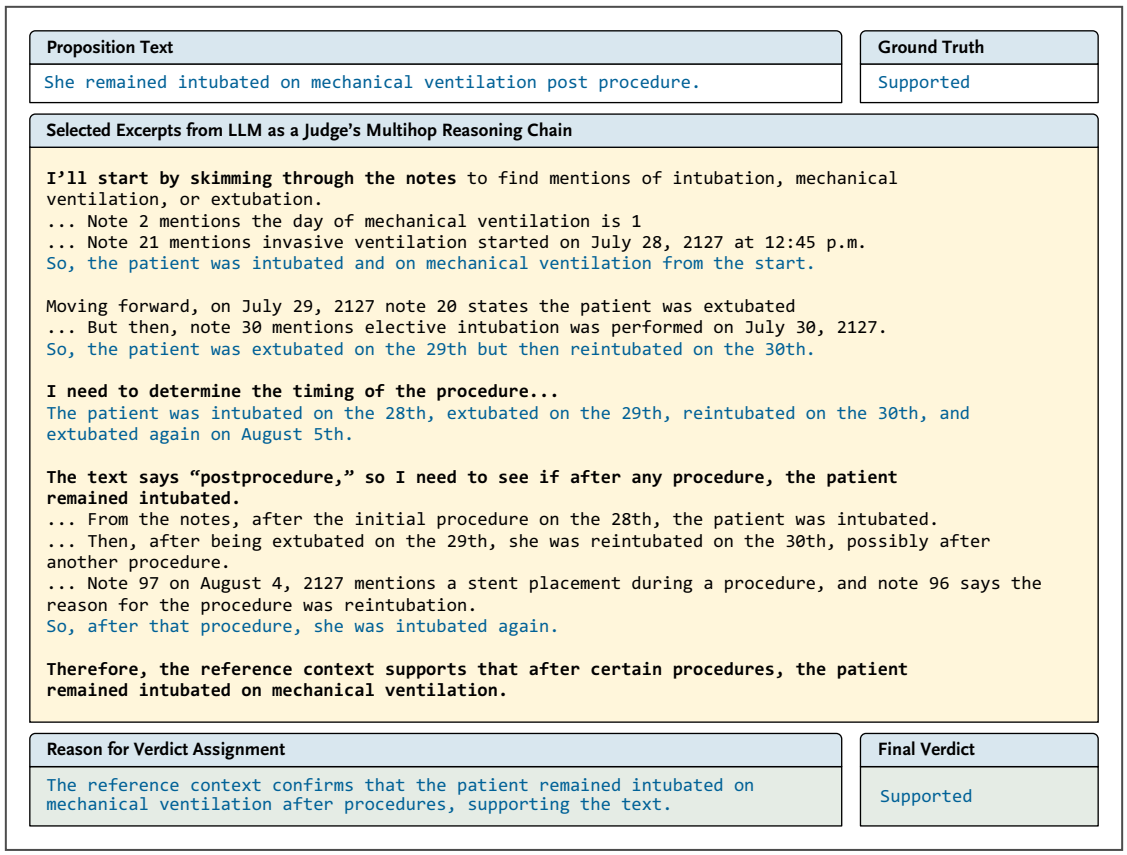


Figure 4. Multihop and Compositional Reasoning in VeriFact.

Select excerpts from DeepSeek-R1-Distill-Llama-70B's reasoning chain that illustrate human-like multihop reasoning and compositional reasoning. The bolded text in the reasoning chain illustrates distinct phases in logical reasoning where DeepSeek-R1-Distill-Llama-70B breaks the overall task of verifying the proposition against facts retrieved from the electronic health record into smaller subtasks. Blue text represents intermediate conclusions in the reasoning chain. DeepSeek-R1-Distill-Llama-70B first scans the notes to isolate key events. Then DeepSeek-R1-Distill-Llama-70B recognizes that the timing of intubation and extubation events is important and summarizes the temporal relationship between these key events. Finally, DeepSeek-R1-Distill-Llama-70B relates the timing of key events with the proposition text and performs compositional reasoning to arrive at the correct answer. The full reasoning chain with the reference context of electronic health record facts that the large language model as an evaluator of large language model output is referencing is shown in Figure S10. LLM-as-a-Judge denotes large language model as an evaluator of large language model output.

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parameter count and addition of reasoning behavior independently improve model performance (Fig. S10). When increasing model size, Llama-3.1-70B-Instruct achieves 7.1% higher agreement than Llama-3.1-8B-Instruct, and DeepSeek-R1-Distill-Llama-70B achieves 6.5% higher agreement than DeepSeek-R1-Distill-Llama-8B (Fig. S11). When adding reasoning behavior, DeepSeek-R1-Distill-Llama-8B achieves 5.1% higher agreement than Llama-3.1-8B-Instruct, and DeepSeek-R1-Distill-Llama-70B achieves 4.3% higher agreement than Llama-3.1-70B-Instruct. Increasing model size improves the LLM-as-a-Judge's clinical interpretation of facts, whereas reasoning yields improvement by breaking down tasks into smaller steps. In Figure S12, DeepSeek-R1-Distill-Llama-8B's reasoning makes verbatim interpretations of EHR facts and is misaligned with the clinician consensus ground truth; in contrast, DeepSeek-R1-Distill-Llama-70B is able to reason and interpret the facts more similarly to clinicians and compose evidence to arrive at the correct conclusion. Smaller nonreasoning models such as Llama-3.1-8B-Instruct have a U-shaped performance in Figure 3, indicating worse performance when too many facts are retrieved, but this behavior can be overcome with the addition of reasoning, an increase in model size, or both (Fig. S13).

Discussion

Patient-specific fact verification of LLM-generated text can be automated using VeriFact, with over 93% agreement with consensus clinician chart review using locally hosted, open-source AI models without task-specific or medical-specific training (Figs. 3 and 4). VeriFact's judgments are more consistent than those of individual human clinicians performing the same task, as the highest clinician inter-rater agreement is 88.5% (Table 1).

This study specifically examines VeriFact's fact-verification capability on LLM-written clinical notes, but other applicable use cases include referral letters, letters to insurance companies, patient instructions, and provider handoff summaries. In these scenarios, VeriFact can direct clinicians to review factually inaccurate statements in an LLM-drafted document so that clinicians can correct them before the document is finalized. In addition, AI-based documentation generation systems can also use VeriFact as a quality-control tool, targeting a specific percentage of supported facts before presenting to a clinician.

Experiments support atomic claim propositions as a unit of evaluation over sentences to reduce the ambiguity of what is being evaluated. This is reflected by higher clinician inter-rater agreement with atomic claim propositions (Table 1) and by LLM judges achieving higher MCCs with atomic claim propositions (Fig. 3). As there is an average of 2.2 atomic claims per sentence, atomic claim propositions also allow for more fine-grained evaluation. However, the optimal fact representation for EHR reference documents is sentences, with atomic claim representations consistently showing degraded performance (Fig. 3). This finding deviates from prior research[38] and suggests errors or information loss when EHR documents are transformed into atomic claims. For example, sentences may contain important relationships between clinical facts that then become isolated into separate atomic claims; the retrieval process may not extract the whole set of related concepts. In this scenario, information retrieved by VeriFact is no longer equivalent to the original sentence. Studies have shown that different LLMs have varying capabilities for producing atomic and lossless claim decompositions.[22-25,54] Further research is needed to investigate whether information retrieval using atomic claims is possible without information loss.

The best LLM-as-a-Judge was DeepSeek-R1-Distill-Llama-70B, as it achieved both high agreement with clinicians and MCCs. Increasing LLM size resulted in improved accuracy due to models' greater ability to attend to relevant facts in the reference context and greater ability to interpret the clinical meaning underlying the facts. Smaller models with fewer model parameters and attention heads, like Llama-3.1-8B-Instruct, may fail to attend to relevant EHR facts as the number of facts increases.[55,56] Reasoning models (e.g., DeepSeek-R1-Distill-Llama-8B, DeepSeek-R1-Distill-Llama-70B, and Qwen-3-30B-A3B-Thinking) generate intermediate "thinking" tokens for multihop reasoning over EHR facts before yielding a final answer (Fig. 4, Figs. S10, S12, and S13). Although reasoning improves alignment with clinician labels (Fig. 3), the additional thinking tokens increase latency in VeriFact and may impede the usability of the system as a fact-verification tool.

Our study has several limitations. First, additional retrieval and reranking models were not explored. Second, only general domain models were studied; no medicine-specific models were included, and no domain-specific fine-tuning was performed. Third, a fixed set of prompts was used for each task and further prompt optimization may be possible. Fourth, VeriFact uses the EHR as the source of truth and, thus, may be inaccurate for patients who are new to a health system, due to a dearth of records, even if the information is known to patients and clinicians. EHRs have also been noted to contain erroneous information

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due to misdiagnosis, miscommunication, or the copying and pasting of outdated information.[57] Thus, VeriFact's accuracy and utility are limited by how well clinicians collectively curate the information within each patient's EHR. Fifth, VeriFact can only evaluate information stated in LLM-generated text and does not address errors of omission. Sixth, VeriFact-BHC contains LLM-generated summaries only from a single LLM and documentation creation pipeline, which may not generalize to other documentation generation systems. In addition, VeriFact-BHC focuses only on verifying facts in discharge summary narratives derived from MIMIC-III, a dataset that has characteristics that limit generalizability: it is a single-center dataset, most patients had one or two admissions, and many inpatient and outpatient notes are omitted. Finally, VeriFact performs well for LLM-written clinical documents, but has significantly decreased performance when evaluating human-written BHC (see the Results section in the Supplemental Appendix). Further research is needed to develop an evaluation system that can perform equally well on human-written clinical documents.

Conclusion

VeriFact can automatically verify facts in LLM-generated clinical documents against patient EHRs with high agreement when compared with human clinicians, making it a promising proxy for manual clinician evaluation.

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Disclosures

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